

Lab-2 Ex7 - Splitting the Bill (II)

In Lab-2 Ex3 you have created a simple program to split the bill. Let's create a more advanced program that considers children!

Let's assume that when you split the bill, children only need to pay half price of adults. For instance, if a \$100 meal is shared by 2 adults and 1 child:

- You can consider a child as 0.5 of an adult; so in total you can consider the total number of people is 2.5
- The adults will thus pay $\$100 / 2.5 = \40 while the child will pay $\$40 * 0.5 = \20 .

The improved C program will ask for:

- How many **adults** enjoyed the meal (always > 0)
- How many **children** enjoyed the meal (for simplicity, always > 0)
- The total **cost** of the meal (non-negative)
- The **tips** (non-negative)

Assume everyone will share both the cost of the meal and the tips. The program should output ***how much each adult and each child should pay***, separately, in 2 decimal places.

Assuming all input are valid, here is how the program should run:

Sample Run #1 (user input is bolded and in italic)

Input number of adults: **2**

Input number of children: **1**

Input cost of meal: **95.5**

Input tips: **4.5**

Each adult pays: 40.00

Each child pays: 20.00

Sample Run #2 (user input is bolded and in italic)

Input number of adults: **5**

Input number of children: **2**

Input cost of meal: **200**

Input tips: **20**

Each adult pays: 36.67

Each child pays: 18.33